

ARNAV SALU

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EDUCATION

University of California, Davis Davis, CA

B.S. Electrical Engineering, Minor in Computer Science Sep 2024 – Jun 2028

Relevant Coursework: Digital Logic Design, Circuit Analysis, Signals & Systems, Embedded Systems, Data Structures

TECHNICAL SKILLS

Languages: C, C++, Python, MATLAB, JavaScript, Verilog, MIPS Assembly

Hardware & Embedded: Arduino, Raspberry Pi, OpenBCI Cyton Board, Intel DE10-Lite FPGA, Breadboard/PCB Prototyping, IoT Sensors, I2C, SPI, UART, Oscilloscopes, DMMs

Tools & Frameworks: Quartus Prime, ModelSim, KiCad, LTspice, MATLAB/Simulink, OpenVIBE, PsychoPy, ThingSpeak, Git, Linux, EDA Playground

ML & Signal Processing: SVM, LDA, RMS Feature Extraction, MVC Normalization, Digital Filtering, EEG/EMG Pipelines, RTL Synthesis, CPU Architecture

Other: PCB Design & Schematic Capture, CAD (Onshape), Technical Documentation, Signal Verification

EXPERIENCE

SAGE Lab, UC Davis Health (Research Technician) | Python, IoT Sensors, Raspberry Pi Nov 2025 – Present

- Automated data ingestion and synchronization from smart-home IoT sensor networks and Raspberry Pi microcontrollers, enabling continuous behavioral data collection for health-monitoring research across participant homes.
- Diagnosed and resolved Bluetooth/Wi-Fi connectivity failures and data sync issues across embedded and networked devices; automated firmware update workflows to reduce manual maintenance.
- Managed power systems and on-site device maintenance to sustain reliable multi-device operational scripts across a distributed, real-world sensor deployment.

Neurotech@Davis (Hardware Team Member) | OpenBCI, MATLAB, Arduino, SVM/LDA Oct 2024 – Present

- Designed and prototyped brain-computer interface hardware linking neural and muscular signals to digital control systems, from wearable biosignal acquisition to real-time embedded actuation.
- Built signal acquisition pipelines on the OpenBCI Cyton Board, applying digital filtering, RMS feature extraction, MVC normalization, and SVM/LDA classification for multimodal neural and EMG control.
- Integrated EEG (SSVEP, Alpha) and EMG signal streams with Arduino hardware via MATLAB serial communication, enabling 6 distinct real-time control commands for a prototype RC car.

UC Irvine, COSMOS Program (Research Student) | Python, Astrophysics Jul 2023 – Aug 2023

- Studied universal expansion and particle physics under UCI faculty; derived and analyzed theoretical energy thresholds for quantum gravity effects using Fermi Gamma-ray Space Telescope data.
- Presented original research on quantum gravity effects on gamma-ray burst photon emissions to UCI graduate students, using Python to extract high-energy photon events and construct per-GRB light curves.

PROJECTS

OpenMed — Biomedical Instrumentation Platform | KiCad, LTspice, Python Jun 2026 – Present

- Designing a 2-channel differential EMG acquisition platform, including the analog front end, instrumentation amplification, filtering, ADC interface, and 2-layer KiCad PCB architecture.
- Evaluating semi-custom and dedicated biomedical AFE architectures via LTspice simulation, quantitatively characterizing gain, frequency response, noise, CMRR, and ADC performance.
- Building an embedded USB acquisition and Python signal-processing pipeline for real-time visualization, filtering, RMS/envelope extraction, and experimental data logging.

Custom 8-Bit RISC CPU on Intel DE10-Lite FPGA | Verilog HDL, Intel Quartus Prime, Icarus Verilog Jun 2026 – August 2026

- Architected and synthesized an 8-bit single-cycle processor featuring a custom ISA, 4-register dual-port register file, ALU, and dedicated control unit on an Altera MAX 10 FPGA.
- Derived unit logic and full system datapath using Icarus Verilog testbenches and EPWave waveforms, validating clock cycles and register write-backs ($5 + 3 = 8$).
- Resolved edge-case hardware bugs including instruction aliasing and PC free-running display flicker, establishing steady execution outputs mapped to 7-segment and LED displays.

Behavioral Deviation Detection (Hearth) | Python, scikit-learn, Full-Stack Mar 2026 – July 2026

- Engineered daily behavioral features from 3.7M smart-home sensor events and trained 5 supervised models using chronological splitting; the SVM achieved an F1 score of 0.67 and an ROC-AUC of 0.88.
- Validated deviation labels against unsupervised methods, confirming flagged days are structurally distinguishable without ground truth.
- Built and deployed Hearth, a full-stack companion app that translates model output into plain-language daily status, with live SMTP email notifications, persistent storage, and Render deployment.